

Role Consistency in Role-Playing Agents: A Survey of Evaluation, Memory, Reasoning, Alignment, and Safety

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Abstract

Large language models have made role-playing agents plausible as interactive characters, virtual companions, game agents, and social actors. Yet the key challenge is not only imitating a persona in one response, but preserving a coherent role across long, dynamic, and adversarial interactions. We survey recent work on *role consistency*: maintaining stable identity, character knowledge, expressive style, memory, values, behavioral logic, and safety boundaries over time. We organize the literature around role specification, role-state maintenance, role-aware reasoning, role-consistent alignment, and evaluation.

1 Introduction

Role-playing agents (RPAs) have become one of the most visible and demanding applications of large language models (LLMs). In fiction-oriented systems, they are expected to revive characters with recognizable personalities and memories. In games and virtual worlds, they must behave as non-player characters with persistent goals and social relations. In education, counseling, training, and social simulation, they are asked to enact patients, clients, students, citizens, historical figures, or synthetic users. Across these settings, the quality of an RPA depends not only on fluency or helpfulness, but on whether the agent remains recognizably itself during sustained interaction.

This requirement exposes a gap between ordinary dialogue generation and role-playing. A model may successfully answer in the style of a character for one or two turns, yet gradually drift into a generic assistant, forget prior events, accept false claims about its background, change its emotional stance, or violate the behavioral boundaries of the assigned role. Such failures are especially likely in long conversations, multi-scenario interactions, adversarial questioning, and environments

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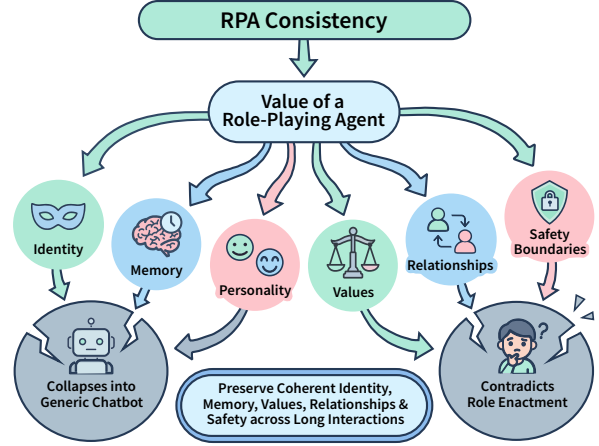


Figure 1: **Why RPA consistency matters.** A trustworthy role-playing agent must preserve coherent identity, long-term memory, personality and values, relationships, and safety boundaries across long, changing, personalized, and adversarial interactions; otherwise it risks collapsing into a generic chatbot or violating the intended role.

where user instructions, retrieved memories, safety policies, and role constraints compete. We refer to the ability to resist these failures as *role consistency*.

Role consistency is broader than persona consistency. Persona consistency has traditionally focused on whether generated responses agree with a set of persona facts or match a character profile. This remains important, but current RPAs require a more dynamic notion. A role includes not only static attributes such as name, occupation, personality, and speaking style, but also situated identity, goals, relationships, knowledge boundaries, psychological orientation, values, and action policies. Thus, role consistency is a longitudinal property of an interactive agent: it concerns whether the agent preserves a coherent role state as the conversation and environment evolve.

Recent research has begun to attack this problem from several directions. Benchmarking work



Figure 2: **Keyword hotness** in 100 recent arXiv papers related to role-playing agents.

evaluates role-playing ability, personality fidelity, character knowledge, and multi-turn consistency (Wang et al., 2024a; Tu et al., 2024; Wang et al., 2024b, 2025b; Samuel et al., 2025). Memory and retrieval methods attempt to preserve long-term facts, experiences, and relationships (Li et al., 2023; Wang et al., 2025c; Park et al., 2025; Chen et al., 2025b). Role-aware reasoning methods ask models to deliberate from within a character state rather than merely produce stylistic imitation (Tang et al., 2025; Xu et al., 2025; Wei et al., 2025; Tang et al., 2026b). Training and alignment methods use contrastive learning, preference optimization, reinforcement learning, and verifiable rewards to stabilize character behavior (Song et al., 2020; Shea and Yu, 2023; Ji et al., 2025; Wang et al., 2025d). Meanwhile, safety and boundary-aware work studies how to preserve character fidelity without enabling unsafe, manipulative, or policy-evading behavior (Tang et al., 2024; Li et al., 2026).

2 Related Surveys and Scope

Recent surveys have covered the broader RPA and persona landscape, including data, modeling, agent architectures, alignment, applications, personalization, and evaluation design (Chen et al., 2024b; Wang et al., 2026d; Chen et al., 2024a; Tseng et al., 2024; Chen et al., 2025a). Our scope is narrower and complementary: we focus on *role consistency* as a longitudinal role-state maintenance problem. From this perspective, an RPA must preserve an evolving representation of who it is, what it knows, how it speaks, what it remembers, what it values, and how it should act. Section 3 defines the conceptual boundary between persona, role, personalization, and role consistency, then explains role specification as the initial role state. Section 4 surveys methods for maintaining role state through memory and retrieval, role-aware reasoning, and

training or alignment. Section 5 reviews benchmark families and the consistency dimensions they measure or miss. Section 6 discusses open challenges, and Section 7 concludes.

3 From Persona to Role Consistency

A timeline view. Figure 3 summarizes the conceptual shift behind this survey. Early work can be understood as *profile matching*: a system is given a persona profile and evaluated by whether responses agree with persona facts or fit a broad style. As RPAs move into games, social simulation, virtual companions, counseling, and training, the role must instead be enacted in an unfolding situation.

The next shift is from *role enactment* to *state maintenance*. A role-playing agent (RPA) is assigned an explicit persona, role, or identity and expected to interact according to its background, style, memory, goals, motivations, and constraints. A *persona* is usually a static profile; a *role* further includes situated identity, social relations, knowledge boundaries, and behavioral constraints. Role consistency therefore concerns whether the agent preserves an evolving role state, not only whether it sounds in character.

We define *role consistency* as the ability of an RPA to maintain coherent identity, knowledge boundaries, style, memory and relations, values, and behavioral logic across multi-turn, multi-scenario, and long-term interaction. This definition follows Figure 3: evaluation expands from fact agreement to role-state fidelity; roles move from profile text into agentic environments; memory moves from local entailment to external stores; control moves from prompting to reasoning and alignment; and stress conditions expand to multimodal, conflicting, and unsafe requests.

The final phase frames RPA consistency as constrained agency. A consistent agent must preserve role fidelity while handling personalization, task goals, platform policies, and safety constraints. Because these objectives can conflict, the rest of the survey treats role consistency as a systems problem spanning role specification, role-state maintenance, role-aware reasoning, role-consistent alignment, and evaluation.

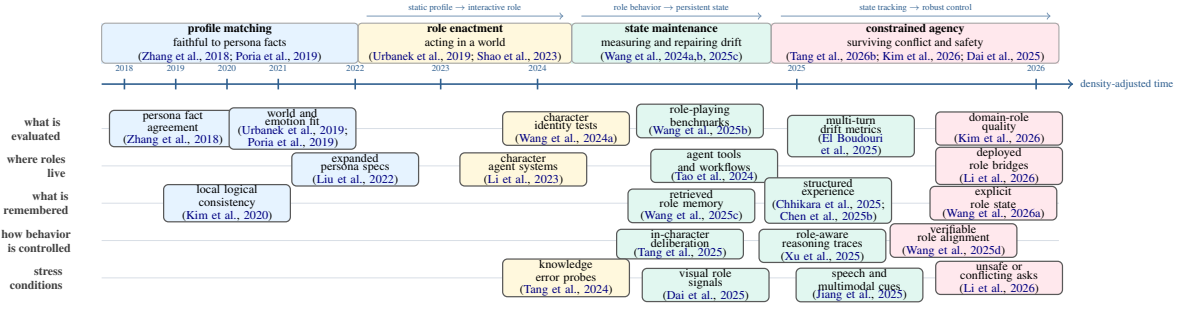


Figure 3: **A structured timeline of role-playing-agent consistency research.** The field has moved from checking whether a response matches static persona facts, to enacting roles in interactive systems, to maintaining measurable role state, and finally to preserving role fidelity under reasoning, alignment, safety, and multimodal stress.

Concept: From Profile to State

The central shift is from checking whether responses match a static persona profile to maintaining an evolving role state. Role consistency therefore depends on what the agent knows, remembers, values, refuses, and is allowed to infer as interaction unfolds.

Role Specification as Initial State. Role specification is the entry point of RPA consistency because it defines the initial role state and determines what should remain stable during interaction. Early persona-consistent dialogue research often represents a role as a small set of natural-language persona facts and evaluates whether responses contradict them (Song et al., 2020; Mesgar et al., 2021). Such flat profiles cover only a narrow part of role state. As RPAs move from profile matching to role enactment, the role no longer lives only in a prompt or character card, but also in scenes, scripts, relationships, virtual worlds, and behavioral constraints (Shanahan et al., 2023; Shao et al., 2023; Li et al., 2023; Yang et al., 2024). A richer specification may therefore include biography, style, traits, relations, private knowledge, goals, values, scenario assumptions, and role boundaries (Ran et al., 2024; Kim et al., 2023; Tang et al., 2026a).

From a role-state perspective, specification should also constrain memory, reasoning, and alignment. The system must distinguish canonical facts, interaction-derived memories, inferred traits, and information outside the role’s knowledge boundary; it should also define when the role should refuse, which goals take priority, and how safety constraints interact with role fidelity (Park et al., 2025; Chen et al., 2025b; Cheng et al., 2025; Tang et al., 2024; Li et al., 2026). Persona expansion, commonsense augmentation, and dynamic refinement can fill gaps in the initial state, but the central challenge is controlled completion: enriching the role without overriding explicit facts or violat-

ing knowledge and safety boundaries (Liu et al., 2022; Kim et al., 2023; Yao et al., 2025; Yang et al., 2026a). The goal is therefore not a longer character card, but an executable, updateable, and auditable initial role state.

4 Method Landscape

We group method-oriented work into three primary categories after role specification has been defined in Section 3: memory and retrieval, role-aware reasoning, and training/alignment. This grouping follows the functional role each method plays in maintaining consistency. Memory and retrieval methods preserve role state over time; reasoning methods infer what the role should know, feel, decide, or refuse in new situations; training and alignment methods make role-faithful behavior more reliable under competing objectives. We discuss benchmarks and evaluation datasets separately in Section 5. Persona expansion and control mainly support role specification, while system frameworks integrate several method categories and are therefore discussed as supporting evidence rather than independent taxonomy axes. Figure 4 summarizes the three functional mechanisms, and Figure 5 attaches representative papers to each leaf category.

4.1 Memory and Retrieval

As shown in Figure 4, memory methods update the role state by filtering new interaction evidence through role boundaries and retrieving relevant prior memory.

Role-state memory. Memory and retrieval work addresses the temporal dimension of role consistency. Without memory, an agent may preserve a role’s surface style while losing continuity in facts, relationships, and experiences. Recent work shows

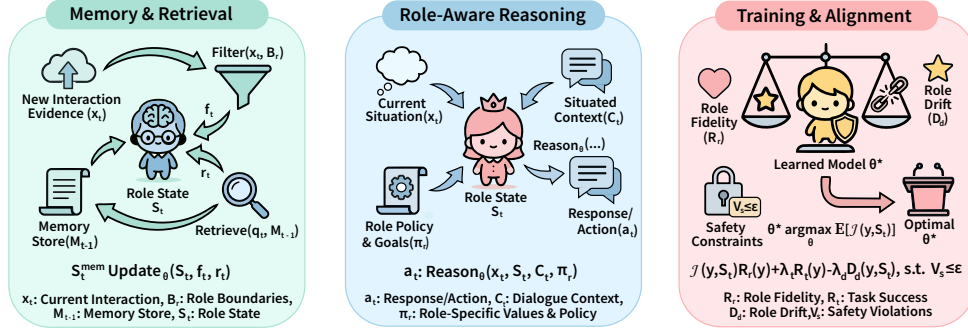


Figure 4: **Three mechanisms for maintaining role consistency.** Memory preserves role state, reasoning grounds role-aware actions, and alignment stabilizes behavior under task and safety constraints.

a progression from script- or context-based memory toward richer long-term role states. ChatHaruhi uses script memory to revive anime characters through LLM-based interaction (Li et al., 2023). MOOM studies maintenance, organization, and optimization of memory in ultra-long role-playing dialogues (Chen et al., 2025b). LightMem separates lightweight online retrieval from offline memory consolidation with small language models, illustrating the efficiency trade-off in long-term agent memory (Zhang et al., 2026). Related work further explores explicit role-state memory, multimodal persona memory, and psychologically motivated memory representations (Wang et al., 2026a; Nie et al., 2026; Cheng et al., 2025).

Retrieval grounding. Retrieval methods ground role enactment in external character knowledge, dialogue history, or interaction traces. RoleRAG introduces graph-guided retrieval to support role-playing with structured character knowledge (Wang et al., 2025c), while AMADEUS adapts retrieval context dynamically for consistent role-playing (Park et al., 2025). Broader memory systems such as Mem0 show how scalable long-term memory can support production agents, although not all such systems are designed specifically for character fidelity (Chhikara et al., 2025). HASH-RAG combines deep hashing with fine-grained retrieval for efficient retrieval-augmented generation, providing a complementary systems perspective rather than an RPA-specific method (Guo et al., 2025). TailorRPA, knowledge-graph conversation, triplet-based RAG, commonsense persona expansion, and CharacterRAG-style resources further indicate that retrieval is becoming a general mechanism for eliciting coherent and personalized RPAs (Gao et al., 2025; Clay and Jiménez-Ruiz, 2024; Wang et al., 2026c; Kim et al., 2023).

Memory control. The common problem across this category is not simply how to store more information, but how to store and retrieve information from the role’s perspective. More memory can improve factual continuity, but it can also introduce irrelevant details, contradictions, privacy risks, and role-boundary violations. For RPAs, memory should preserve what the character should know, remember, and infer, not everything the underlying system can access. This makes memory control, provenance, temporal ordering, context compression, and knowledge-boundary filtering central to role consistency. Dynamic persona refinement, multi-turn consistency probes, training-free context compression, and persona-bridging safety tests show why memory updates must be filtered through role boundaries rather than accepted as raw conversation state (Yao et al., 2025; Kim et al., 2026; Zhou et al., 2026; Yang et al., 2026a). Adjacent dialogue and multi-agent work provides complementary mechanisms for reducing context contamination: DisenCRS disentangles focal information from background dialogue context, while gated co-ordination separates verified private working memory from protocolized public coordination state (An et al., 2025; Jian et al., 2026).

4.2 Role-Aware Reasoning

As shown in Figure 4, reasoning methods use the maintained role state, situated context, and role policy to produce a role-grounded response or action.

Standpoint reasoning. Role-aware reasoning work treats role consistency as a deliberation problem. A model may know the right facts but reason from the wrong standpoint, producing responses that are useful as an assistant but implausible for the assigned role. Thinking in Character explicitly targets role-aware reasoning (Tang et al., 2025),

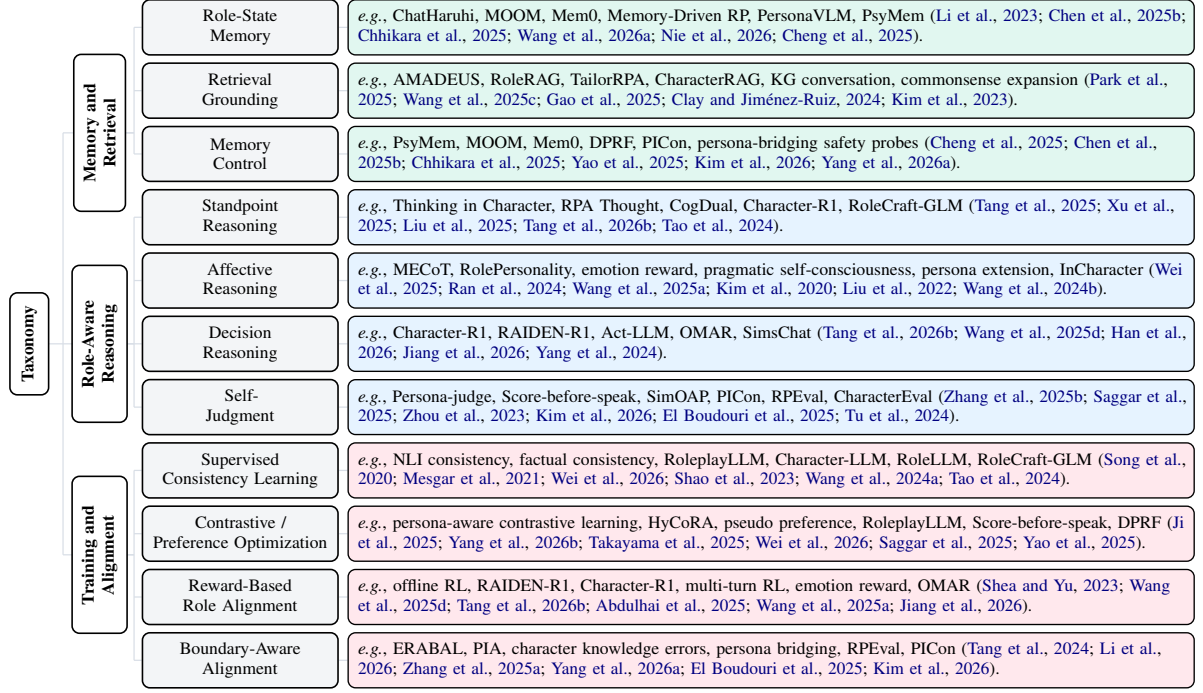


Figure 5: **Taxonomy of method-oriented work for role consistency in role-playing agents.** The rightmost boxes list representative papers and can be extended as the literature grows.

while RPA Thought evaluates inner-thought reasoning for role-playing language agents (Xu et al., 2025). Prompt-level methods that improve chain-of-thought behavior or resolve prompt ambiguity are relevant adjacent mechanisms because they affect whether the model interprets the role and situation consistently (Chen et al., 2026; Huang et al., 2026). CogDual, Character-R1, and RoleCraft-GLM similarly frame role fidelity as a question of reasoning from role-specific cognition, constraints, or workflows (Liu et al., 2025; Tang et al., 2026b; Tao et al., 2024).

Affective reasoning. Many role failures are emotional or motivational rather than purely factual. MECoT introduces emotional chain-of-thought for personality-consistent role-playing (Wei et al., 2025). Related work on role personality, emotion rewards, pragmatic self-consciousness, persona extension, and psychological interview evaluation further suggests that role consistency requires models to represent emotions, values, and social intentions as active parts of deliberation (Ran et al., 2024; Wang et al., 2025a; Kim et al., 2020; Liu et al., 2022; Wang et al., 2024b).

Decision reasoning. This category moves beyond surface imitation by asking what the character would choose, refuse, prioritize, remember, or do in a new situation. Character-R1 and RAIDEN-

R1 use verifiable rewards to strengthen role awareness through RL-style training (Tang et al., 2026b; Wang et al., 2025d). Related role-aware reasoning work points toward role-specific decision rules grounded in goals, constraints, and action policies, including tool-using agents, multi-agent role play, collaborative prompt optimization, and simulated social interaction (Han et al., 2026; Jiang et al., 2026; Wang et al., 2026b; Yang et al., 2024). Weak-link optimization further suggests that consistency can be distributed across coordinated agents rather than confined to a single model state (Bian et al., 2026).

Self-judgment. Self-judgment methods evaluate whether intermediate thoughts or final responses remain compatible with the role. Persona-judge studies token-level self-judgment for personalized alignment (Zhang et al., 2025b). Score-before-speak, SimOAP, PICon, RPEval, and CharacterEval illustrate adjacent uses of scoring, probing, or judging to detect whether role behavior remains consistent before or after generation (Saggar et al., 2025; Zhou et al., 2023; Kim et al., 2026; El Boudouri et al., 2025; Tu et al., 2024). The main risk is that explicit reasoning traces can become performative: they may reveal hidden instructions, rationalize inconsistent behavior, or optimize for judge-visible explanations rather than genuine role

fidelity. Future work needs reasoning protocols that improve role-grounded decisions while remaining evaluable, safe, and compatible with role boundaries.

4.3 Training and Alignment

As shown in Figure 4, training and alignment methods optimize role-faithful behavior under task and safety constraints rather than maximizing character imitation alone.

Supervised consistency learning. Training and alignment work attempts to make role-consistent behavior more reliable than prompting alone. Earlier persona-consistent dialogue work uses natural language inference and factual consistency objectives to reduce contradictions between responses and persona facts (Song et al., 2020). Factual-consistency training, Character-LLM, RoleLLM, RoleplayLLM, and RoleCraft-GLM show how supervised or instruction-style data can ground role behavior before more elaborate preference or reward optimization is applied (Mesgar et al., 2021; Shao et al., 2023; Wang et al., 2024a; Wei et al., 2026; Tao et al., 2024). Supervised data, demonstrations, and scripted role interactions provide the first layer of behavioral grounding.

Contrastive/preference optimization. Contrastive and preference-based methods compare more and less role-consistent behaviors. Persona-aware contrastive learning, HyCoRA, and multidimensional fine-tuning shape model representations and preferences toward character fidelity (Ji et al., 2025; Yang et al., 2026b; Wei et al., 2026). Offline reinforcement learning has been used to build persona-consistent dialogue agents (Shea and Yu, 2023), while pseudo preference optimization and response scoring provide additional routes for improving persona consistency (Takayama et al., 2025; Saggar et al., 2025).

Reward-based role alignment. RL-based approaches such as RAIDEN-R1, Character-R1, verifiable emotion rewards, and multi-turn self-play methods show a newer trend: using reward signals to stabilize role-aware behavior across longer and more interactive settings (Wang et al., 2025d; Tang et al., 2026b; Wang et al., 2025a; Abdulhai et al., 2025). OMAR and related multi-agent or role-reward settings further suggest that rewards can target decisions and interactions rather than only single responses (Jiang et al., 2026). These meth-

ods are especially relevant when role consistency cannot be reduced to exact-match facts and must be learned from comparative or verifiable signals.

Boundary-aware alignment. The alignment challenge is that role consistency is not the only objective. A deployed RPA must also follow user instructions, satisfy task goals, obey safety policies, and avoid harmful behavior. Boundary-aware learning such as ERABAL connects role fidelity with safety constraints (Tang et al., 2024), while persona-invariant safety alignment studies how adversarial role prompts can pressure models to violate safety policies (Li et al., 2026). Character knowledge-error detection, persona bridging, RPEval, and PICon further show that boundary failures can appear as factual overreach, role drift, or unsafe compliance under pressure (Zhang et al., 2025a; Yang et al., 2026a; El Boudouri et al., 2025; Kim et al., 2026). Thus, alignment for RPAs is inherently conflict-aware: methods must specify when role fidelity should dominate, when safety or user intent should override it, and how refusals can remain compatible with the role rather than collapsing into generic assistant behavior.

💡 Insight: Layered Maintenance

Role consistency is not solved by a single technique, but by a layered maintenance process: specification defines the role, memory preserves it, reasoning makes the model act from the role’s standpoint, and alignment stabilizes role behavior under competing objectives.

5 Evaluation Practices

Evaluation is the bottleneck of RPA consistency. Because role consistency spans identity, knowledge, expression, memory, values, decisions, and safety boundaries, no single benchmark can fully measure it. We summarize current evaluation practices according to the evidence they collect and the risks they leave unresolved.

5.1 Benchmark Map

Table 1 is the organizing device for this section. It groups representative benchmarks by task setting and marks six consistency dimensions: knowledge/factual consistency (K), expression/style consistency (E), psychological/personality fidelity (P), memory and temporal continuity (M), role-aware reasoning and decision consistency (R), and robustness to role drift or boundary pressure (B). The table should be read as a benchmark card rather than a leaderboard: each row records what kind of evi-

Benchmark-card schema. Rows are grouped by task setting; metadata columns describe source and scale, while the middle block marks six role-consistency dimensions (✓ primary, △ partial, − weak) using author-side coverage annotations. K = knowledge/factual consistency, E = expression/style consistency, P = psychological/personality fidelity, M = memory and temporal continuity, R = role-aware reasoning and decision consistency, and B = robustness to role drift or boundary pressure.

Benchmark	Benchmark metadata				Consistency dimensions						Dataset scale and source	
	Year	Venue	Modality	Lang.	K	E	P	M	R	B	Scale / Source	
(A) Textual Role-Playing <i>Persona/character dialogue and role customization.</i>												
PersonaChat (Zhang et al., 2018)	2018	ACL	Text	En	✓	△	△	−	−	−	10K dial.; 164K utt.; self-described persona about 1K dialogues; hist./literary figures 168K samples; fictional roles 11K samples; novels and scripts 23K annotations; multi-source not reported; open-ended RP	
Character-LLM (Shao et al., 2023)	2023	EMNLP	Text	En	✓	✓	△	△	△	−		
RoleLLM (Wang et al., 2024a)	2024	ACL	Text	En	✓	✓	△	△	△	△		
CharacterEval (Tu et al., 2024)	2024	ACL	Text	Zh	✓	✓	△	△	−	△		
CharacterBench (Zhou et al., 2025)	2025	AAAI	Text	Zh/En	✓	✓	△	△	△	△		
PingPong (Gusev, 2024)	2024	preprint	Text	En	△	✓	△	−	△	△		
(B) Situated Role-Playing <i>Scene- or world-grounded interaction.</i>												
LIGHT (Urbanek et al., 2019)	2019	EMNLP	Text	En	△	✓	△	△	△	−	11K scenes; 21K dial.; fantasy RPG scenes scenario-driven interactions; virtual worlds	
CharacterBox (Wang et al., 2025b)	2025	NAACL	Text	En	△	✓	△	△	△	△		
(C) Diagnostic Probing <i>Questioning, interviews, reasoning, and drift probes.</i>												
InCharacter (Wang et al., 2024b)	2024	ACL	Text	En	△	△	✓	−	△	−	7.6K questions; fictional characters large scale; multi-source roles multi-turn consistency probes; persona profiles thought-reasoning questions; RPA cases counselor dialogue cases; counseling role	
RoleMRC (Lu et al., 2025)	2025	ACL	Text	En	✓	△	−	−	✓	△		
PICon (Kim et al., 2026)	2026	preprint	Text	mixed	✓	△	△	✓	△	✓		
RPA Thought (Xu et al., 2025)	2025	preprint	Text	mixed	△	−	△	−	✓	△		
Counselor (Rudolph et al., 2026)	2026	preprint	Text	mixed	△	△	△	△	✓	✓		
(D) Multimodal Role-Playing <i>Image, video, audio, or speech grounded RP.</i>												
MMRole (Dai et al., 2025)	2025	ICLR	Image Text	En	✓	✓	△	△	△	△	11K images; 294K tests; virtual/hist. roles 60K videos; 700K dial.; daily scenes MMRole-style data; MMRole-derived 117K speech dial.; speech dialogues 261 films; 65.6 hours; film source	
Video2Roleplay (Zhang et al., 2025c)	2025	EMNLP	Video Text	Zh	✓	✓	△	✓	△	△		
MMRole-Agent (Dai et al., 2025)	2025	ICLR	Image Text	mixed	✓	✓	△	△	△	△		
SpeechRole (Jiang et al., 2025)	2025	preprint	Speech	En	△	✓	✓	△	△	△		
VoxRole (Wu et al., 2025)	2025	preprint	Speech	En	△	✓	△	△	△	△		
(E) Adaptable Dialogue Resources <i>Emotion/dialogue resources adaptable to RP evaluation.</i>												
MELD (Poria et al., 2019)	2019	ACL	Video Audio Text	En	−	△	✓	△	−	−	1,433 dial.; 13.7K utt.; Friends 10K utt.; 12 hours; lab acting	
IEMOCAP (Busso et al., 2008)	2008	dataset	Audio Video Text	En	−	△	✓	△	−	−		

Table 1: A benchmark metadata framework with representative samples grouped by task setting.

dence a benchmark can provide, while the empty or partial cells indicate what remains under-measured. The primary/partial/weak labels are our author-side annotations based on each benchmark’s task design, data source, reported metrics, and evaluation protocol; they are intended as coverage indicators rather than claims of absolute benchmark quality.

Textual Role-Playing. The first block of Table 1 contains the core textual benchmarks. PersonaChat represents the earlier persona-dialogue setting, where consistency is mostly grounded in agreement with self-described persona facts (Zhang et al., 2018). Character-LLM, RoleLLM, CharacterEval, CharacterBench, and PingPong broaden this setting toward richer character profiles, stylistic imitation, and open-ended role-playing (Shao et al., 2023; Wang et al., 2024a; Tu et al., 2024; Zhou et al., 2025; Gusev, 2024). Their strongest coverage is usually K and E: they can detect whether a model knows the character and whether it sounds plausibly in character. Their weaker coverage is M, R, and B. A model may answer character facts correctly and mimic style while still failing over long interaction, accepting false user claims, or making

decisions that contradict the role.

Situated Role-Playing. The second block shifts evaluation from profile matching to scene- or world-grounded interaction. LIGHT and CharacterBox evaluate whether agents can act within a social or virtual environment rather than merely answer profile questions (Urbanek et al., 2019; Wang et al., 2025b). This setting makes expression, scenario fit, and interactional behavior more visible. It also begins to expose role-state problems: the agent must track where it is, who it is speaking to, and what the situation permits. However, most situated benchmarks still provide only partial evidence for memory continuity and robustness. They test whether a response fits a scene, but not always whether the role remains stable across repeated scenes, delayed references, or adversarial pressure.

Diagnostic Probing. The third block isolates specific consistency dimensions through interviews, reading-comprehension-style probes, thought questions, and role-stress tests. InCharacter turns psychological interviews into a personality-fidelity evaluation (Wang et al., 2024b). RoleMRC probes role-specific knowledge (Lu et al., 2025), while

RPA Thought targets role-specific reasoning (Xu et al., 2025). PIcon and counselor-role evaluations add multi-turn pressure, conflict, and domain-specific role constraints (Kim et al., 2026; Rudolph et al., 2026). These benchmarks are valuable because they make hidden failures measurable: wrong standpoint, unstable values, role drift, or unsafe refusal behavior. Their limitation is construct validity. A probe can show that a model gives a plausible explanation or stable questionnaire answer, but this does not guarantee faithful behavior in open-ended interaction.

Multimodal Role-Playing. The fourth block reflects a newer evaluation frontier. MMRole, Video2Roleplay, MMRole-Agent, SpeechRole, and VoxRole test whether role consistency survives when the role is grounded in images, video, audio, or speech (Dai et al., 2025; Zhang et al., 2025c; Jiang et al., 2025; Wu et al., 2025). These benchmarks expand K and E beyond text: visual appearance, voice, scene cues, and audiovisual context become part of the role. They are especially important for embodied agents, games, virtual companions, and social simulation. At the same time, they make evaluation harder because inconsistency may appear across modalities: a model may preserve textual persona while losing voice style, visual identity, emotional tone, or action coherence.

Adaptable Dialogue Resources. The auxiliary block lists resources that were not designed specifically for RPA consistency but can support evaluation. MELD and IEMOCAP provide emotion-rich multimodal dialogue data (Poria et al., 2019; Busso et al., 2008). They are useful for probing expression, affect, and interactional coherence, especially when RPA evaluation needs emotional or paralinguistic evidence. Their limitation is role specificity: emotion labels and acted dialogue do not by themselves define a stable character state, knowledge boundary, or role-specific decision policy. They are therefore best treated as auxiliary resources rather than direct RPA consistency benchmarks.

5.2 Cross-Dimensional Gaps

Reading Table 1 by columns reveals the main gap. Current benchmarks are strongest for K and E, increasingly useful for P and R, and still sparse for M and B. This imbalance matters because many real RPA failures are longitudinal and conflictual rather than single-turn factual errors. Stronger evaluation should therefore combine automatic checks, human

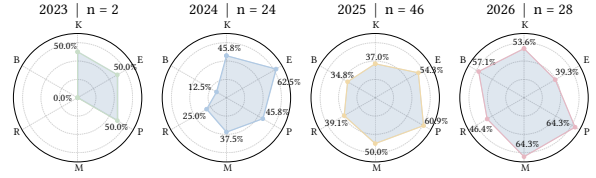


Figure 6: **Temporal change** in role-consistency signals across six dimensions.

judgment, and calibrated LLM-as-judge protocols. Automatic tests can verify facts, knowledge boundaries, and structural constraints; human raters can assess whether behavior feels in-character and emotionally coherent; judges can scale pairwise comparison or rubric scoring, but must be controlled for verbosity bias, familiarity bias, and confusion between helpfulness and role fidelity.

⚠️ Claim: Evaluation Gap

Current evaluation is strongest at testing whether agents know the role and sound in character, but weakest at testing whether they sustain it over time. Linguistic signals make this gap more observable: an agent may preserve persona facts while drifting in politeness strategy, social distance, register, emotional density, or role-specific worldview.

6 Open Challenges and Future Directions

The role-state view highlights several open problems for future NLP research.

Controllable Long-Term Memory. Long-term memory should decide what to remember, forget, update, and reveal. This requires provenance, temporal structure, uncertainty, and boundary control, with clear separation among canon, interaction-derived memory, inferred traits, and user-specific relational memory.

Robust Evaluation Under Role Attacks. Users may induce role drift through false assertions, break-character requests, scenario shifts, or memory manipulation. Evaluation should include role attacks that measure not only refusal, but refusal that remains compatible with the role.

Language-Grounded Multimodal, Embodied, and Agentic Consistency. As RPAs enter games, virtual worlds, social robots, and multimodal assistants, language will mediate perception, intention, social meaning, and action. Future work should study how language anchors voice, gesture, visual identity, tool use, movement, and environmental action to a coherent role state. This shifts evaluation from utterance-level fidelity to language-grounded

role behavior, where inconsistent actions may carry stronger safety consequences than inconsistent text.

7 Conclusion

We surveyed recent work through a compact role-state taxonomy covering role specification, role-state maintenance, role-aware reasoning, and role-consistent alignment, while treating evaluation as a separate cross-cutting problem. The field has made clear progress, but robust RPA consistency will require evaluation and methods that are longitudinal, conflict-aware, and safety-preserving.

Limitations

This survey is based on a rapidly evolving literature on role-playing agents, persona consistency, personalization, agent memory, and safety. Because new preprints and benchmarks appear frequently, the coverage may become incomplete quickly. Our taxonomy emphasizes role consistency and therefore gives less attention to adjacent topics such as general-purpose dialogue systems, social simulation without explicit role fidelity, and personalization systems that primarily optimize user preference rather than character stability. The categorization of papers is also interpretive: many systems combine memory, reasoning, training, and evaluation components, so assigning them to a primary research line inevitably simplifies their contributions. Finally, we focus mainly on text-based role-playing agents, while multimodal, embodied, and voice-based role consistency remain underrepresented in the current synthesis.

Ethical Considerations

This survey does not introduce new datasets, experiments, or human-subject studies. However, the surveyed area raises substantial ethical issues. Role-playing agents can create strong impressions of identity, memory, intimacy, authority, or expertise, which may mislead users if systems are not transparently presented as artificial. Character fidelity can also conflict with safety: an agent that faithfully enacts a harmful, manipulative, extremist, or medically unreliable role may produce unsafe outputs. Conversely, overly aggressive safety filters may erase intended character boundaries or cultural expression. Memory-based RPAs introduce privacy risks because long-term interactions may contain sensitive user information, inferred preferences, or emotional disclosures. Evaluation

datasets based on fictional characters, public figures, or human personas may also raise copyright, consent, and representational concerns. Responsible RPA development should therefore include transparent disclosure, privacy-preserving memory design, consent-aware data collection, culturally sensitive evaluation, and safety mechanisms that constrain role enactment without relying on hidden or inconsistent behavior.

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References

- Marwa Abdulhai, Ryan Cheng, Donovan Clay, Tim Althoff, Sergey Levine, and Natasha Jaques. 2025. [Consistently simulating human personas with multi-turn reinforcement learning](#). *Preprint*, arXiv:2511.00222.
- Guojia An, Jie Zou, Jiwei Wei, Chaoning Zhang, Fuming Sun, and Yang Yang. 2025. [Beyond whole dialogue modeling: Contextual disentanglement for conversational recommendation](#). In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 31–41. Association for Computing Machinery.
- Haoyu Bian, Chaoning Zhang, Jiaquan Zhang, Xingyao Li, Yuanfang Guo, Wei Dong, and Yang Yang. 2026. [Weak-link optimization for multi-agent reasoning and collaboration](#). *Preprint*, arXiv:2604.15972.
- Carlos Busso, Murtaza Bulut, Chi-Chun Lee, Abe Kazemzadeh, Emily Mower, Samuel Kim, Jeanette N. Chang, Sungbok Lee, and Shrikanth S. Narayanan. 2008. [IEMOCAP: Interactive emotional dyadic motion capture database](#). *Language Resources and Evaluation*, 42(4):335–359.
- Chaoran Chen, Bingsheng Yao, Ruishi Zou, Wenye Hua, Weimin Lyu, Toby Jia-Jun Li, and Dakuo Wang. 2025a. [Towards a design guideline for rpa evaluation: A survey of large language model-based role-playing agents](#). In *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics.
- Jiangjie Chen, Xintao Wang, Rui Xu, Siyu Yuan, Yikai Zhang, Wei Shi, Jian Xie, Shuang Li, Ruihan Yang, Tinghui Zhu, Aili Chen, Nianqi Li, Lida Chen, Caiyu Hu, Siye Wu, Scott Ren, Ziquan Fu, and Yanguhua Xiao. 2024a. [From persona to personalization: A survey on role-playing language agents](#). *Transactions on Machine Learning Research*.
- Nuo Chen, Yan Wang, Yang Deng, and Jia Li. 2024b. [The oscars of ai theater: A survey on role-playing with language models](#). *Preprint*, arXiv:2407.11484.

- Shuxu Chen, Yitian Zhou, Jiaquan Zhang, Haoyu Bian, Aming Wu, Sungyoung Lee, Chaoning Zhang, and Hyundong Shin. 2026. [CAP-CoT: Cycle adversarial prompt for improving chain of thoughts in LLM reasoning](#). *Preprint*, arXiv:2604.23270.
- Weishu Chen, Jinyi Tang, Zhouhui Hou, Shihao Han, Mingjie Zhan, Zhiyuan Huang, Delong Liu, Jiawei Guo, Zhicheng Zhao, and Fei Su. 2025b. [Moom: Maintenance, organization and optimization of memory in ultra-long role-playing dialogues](#). *Preprint*, arXiv:2509.11860.
- Xilong Cheng, Yunxiao Qin, Yuting Tan, Zhengnan Li, Ye Wang, Hongjiang Xiao, and Yuan Zhang. 2025. [Psymem: Fine-grained psychological alignment and explicit memory control for advanced role-playing llms](#). *Preprint*, arXiv:2505.12814.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. [Mem0: Building production-ready ai agents with scalable long-term memory](#). *Preprint*, arXiv:2504.19413.
- Alex Clay and Ernesto Jiménez-Ruiz. 2024. [Information for conversation generation: Proposals utilising knowledge graphs](#). *Preprint*, arXiv:2410.16196.
- Yanqi Dai, Huanran Hu, Lei Wang, Shengjie Jin, Xu Chen, and Zhiwu Lu. 2025. [MMRole: A comprehensive framework for developing and evaluating multimodal role-playing agents](#). In *International Conference on Learning Representations*.
- Yassine El Boudouri, Walter Nuninger, Julian Alvarez, and Yvan Peter. 2025. [Role-playing evaluation for large language models](#). *Preprint*, arXiv:2505.13157.
- Zhenpeng Gao, Xiaofen Xing, and Xiangmin Xu. 2025. [Tailorrrpa: A retrieval-based framework for eliciting personalized and coherent role-playing agents in general domain](#). In *Findings of the Association for Computational Linguistics: EMNLP*, pages 5381–5412. Association for Computational Linguistics.
- Jinyu Guo, Xunlei Chen, Qiyang Xia, Zhaokun Wang, Jie Ou, Libo Qin, Shunyu Yao, and Wenhong Tian. 2025. [HASH-RAG: Bridging deep hashing with retriever for efficient, fine retrieval and augmented generation](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 26847–26858, Vienna, Austria. Association for Computational Linguistics.
- Ilya Gusev. 2024. [Pingpong: A benchmark for role-playing language models with user emulation and multi-model evaluation](#). *Preprint*, arXiv:2409.06820.
- Xiaoxu Han, Wanqing Zhao, Ziyu Guan, and Jinye Peng. 2026. [Act-llm: A whole-process chain for character-centric role-playing with large language models](#). *Expert Systems with Applications*, 296:129024.
- Zhenzhen Huang, Chaoning Zhang, Fachrina Dewi Puspitasari, Jiaquan Zhang, Yitian Zhou, Shuxu Chen, and Yang Yang. 2026. [Small language model helps resolve semantic ambiguity of LLM prompt](#). *Preprint*, arXiv:2604.23263.
- Ke Ji, Yixin Lian, Linxu Li, Jingsheng Gao, Weiyuan Li, and Bin Dai. 2025. [Enhancing persona consistency for llms’ role-playing using persona-aware contrastive learning](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 26221–26238. Association for Computational Linguistics.
- Huadong Jian, Chenghao Li, Haoyu Wang, Jiajia Shuai, Jinyu Guo, Yang Yang, and Chaoning Zhang. 2026. [Gated coordination for efficient multi-agent collaboration in Minecraft game](#). *Preprint*, arXiv:2604.18975.
- Bowen Jiang, Taiwei Shi, Ryo Kamoi, Yuan Yuan, Camillo J. Taylor, Longqi Yang, Pei Zhou, and Sihao Chen. 2026. [One model, all roles: Multi-turn, multi-agent self-play reinforcement learning for conversational social intelligence](#). *Preprint*, arXiv:2602.03109.
- Changhao Jiang, Jiajun Sun, Yifei Cao, Jiabao Zhuang, Xinmeng Che, Hui Li, Xiaoran Fan, Ming Zhang, Junjie Ye, Shihan Dou, Zhiheng Xi, Jingqi Tong, Yilong Wu, Baoyu Fan, Tao Ji, Tao Gui, Qi Zhang, and Xuanjing Huang. 2025. [Speechrole: A large-scale dataset and benchmark for evaluating speech role-playing agents](#). *Preprint*, arXiv:2508.02013.
- Donghyun Kim, Youbin Ahn, Wongyu Kim, Chanhee Lee, Kyungchan Lee, Kyong-Ho Lee, Jeonguk Kim, Donghoon Shin, and Yeonsoo Lee. 2023. [Persona expansion with commonsense knowledge for diverse and consistent response generation](#). In *Proceedings of the European Chapter of the Association for Computational Linguistics*, pages 1139–1149. Association for Computational Linguistics.
- Hyunwoo Kim, Byeongchang Kim, and Gunhee Kim. 2020. [Will i sound like me? improving persona consistency in dialogues through pragmatic self-consciousness](#). In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 904–916. Association for Computational Linguistics.
- Minseo Kim, Sujeong Im, Junseong Choi, Junhee Lee, Chaeun Shim, Hwajung Hong, and Edward Choi. 2026. [Picon: A multi-turn interrogation framework for evaluating persona agent consistency](#). *Preprint*, arXiv:2603.25620.
- Cheng Li, Ziang Leng, Chenxi Yan, Junyi Shen, Hao Wang, Weishi Mi, Yaying Fei, Xiaoyang Feng, Song Yan, HaoSheng Wang, Linkang Zhan, Yaokai Jia, Pingyu Wu, and Haozhen Sun. 2023. [Chatharuhi: Reviving anime character in reality via large language model](#). *Preprint*, arXiv:2308.09597.
- Jiajia Li, Xiaoyu Wen, Zhongtian Ma, Shuyue Hu, Qiaosheng Zhang, and Zhen Wang. 2026. [Disentangling intent from role: Adversarial self-play for persona-invariant safety alignment](#). *Preprint*, arXiv:2605.01899.

- Cheng Liu, Yifei Lu, Fanghua Ye, Jian Li, Xingyu Chen, Feiliang Ren, Zhaopeng Tu, and Xiaolong Li. 2025. [CogDual: Enhancing dual cognition of LLMs via reinforcement learning with implicit rule-based rewards](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 27307–27336, Suzhou, China. Association for Computational Linguistics.
- Yifan Liu, Wei Wei, Jiayi Liu, Xianling Mao, Rui Fang, and Dangyang Chen. 2022. [Improving personality consistency in conversation by persona extending](#). In *Proceedings of the ACM International Conference on Information and Knowledge Management*.
- Junru Lu, Jiazheng Li, Guodong Shen, Lin Gui, Siyu An, Yulan He, Di Yin, and Xing Sun. 2025. [Rolemrc: A fine-grained composite benchmark for role-playing and instruction-following](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 21008–21030. Association for Computational Linguistics.
- Mohsen Mesgar, Edwin D. Simpson, and Iryna Gurevych. 2021. [Improving factual consistency between a response and persona facts](#). In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 549–562. Association for Computational Linguistics.
- Chang Nie, Chaoyou Fu, Yifan Zhang, Haihua Yang, and Caifeng Shan. 2026. [Personavlm: Long-term personalized multimodal llms](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Jeiyeon Park, Yongshin Han, Minseop Kim, and Kisu Yang. 2025. [Dynamic context adaptation for consistent role-playing agents with retrieval-augmented generations](#). *Preprint*, arXiv:2508.02016.
- Soujanya Poria, Devamanyu Hazarika, Navonil Majumder, Gautam Naik, Erik Cambria, and Rada Mihalcea. 2019. [Meld: A multimodal multi-party dataset for emotion recognition in conversations](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.
- Yiting Ran, Xintao Wang, Rui Xu, Xinfeng Yuan, Jiaqing Liang, Yanghua Xiao, and Deqing Yang. 2024. [Capturing minds, not just words: Enhancing role-playing language models with personality-indicative data](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 14566–14576, Miami, Florida, USA.
- Eric Rudolph, Natalie Engert, and Jens Albrecht. 2026. [Evaluating role-consistency in LLMs for counselor training](#). *Preprint*, arXiv:2601.08892.
- Arpita Saggat, Jonathan C. Darling, Vania Dimitrova, Duygu Sarikaya, and David C. Hogg. 2025. [Score before you speak: Improving persona consistency in dialogue generation using response quality scores](#). *Preprint*, arXiv:2508.06886.
- Vinay Samuel, Henry Peng Zou, Yue Zhou, Shreyas Chaudhari, Ashwin Kalyan, Tanmay Rajpurohit, Ameet Deshpande, Karthik Narasimhan, and Vishvak Murahari. 2025. [Personagym: Evaluating persona agents and llms](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 6999–7022, Suzhou, China.
- Murray Shanahan, Kyle McDonell, and Laria Reynolds. 2023. [Role play with large language models](#). *Nature*, 623(7987):493–498.
- Yunfan Shao, Linyang Li, Junqi Dai, and Xipeng Qiu. 2023. [Character-llm: A trainable agent for role-playing](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13153–13187, Singapore, Singapore. Association for Computational Linguistics.
- Ryan Shea and Zhou Yu. 2023. [Building persona consistent dialogue agents with offline reinforcement learning](#). In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 1778–1795, Singapore. Association for Computational Linguistics.
- Haoyu Song, Wei-Nan Zhang, Jingwen Hu, and Ting Liu. 2020. [Generating persona consistent dialogues by exploiting natural language inference](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 8878–8885.
- Junya Takayama, Masaya Ohagi, Tomoya Mizumoto, and Katsumasa Yoshikawa. 2025. [Persona-consistent dialogue generation via pseudo preference tuning](#). In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 5507–5514, Abu Dhabi, UAE. Association for Computational Linguistics.
- Wenqiu Tang, Zhen Wan, Takahiro Komamizu, and Ichiro Ide. 2026a. [Facet-level persona control by trait-activated routing with contrastive sae for role-playing llms](#). *Preprint*, arXiv:2602.19157.
- Yihong Tang, Kehai Chen, Xuefeng Bai, Benyou Wang, Zeming Liu, Haifeng Wang, and Min Zhang. 2026b. [Character-r1: Enhancing role-aware reasoning in role-playing agents via rlvr](#). *Preprint*, arXiv:2601.04611.
- Yihong Tang, Kehai Chen, Muyun Yang, Zheng-Yu Niu, Jing Li, Tiejun Zhao, and Min Zhang. 2025. [Thinking in character: Advancing role-playing agents with role-aware reasoning](#). In *Advances in Neural Information Processing Systems*.
- Yihong Tang, Jiao Ou, Che Liu, Fuzheng Zhang, Di Zhao, and Kun Zhang. 2024. [Erabal: Enhancing role-playing agents through boundary-aware learning](#). *Preprint*, arXiv:2409.14710.
- Meiling Tao, Liang Xuechen, Tianyu Shi, Lei Yu, and Yiting Xie. 2024. [Rolecraft-glm: Advancing personalized role-playing in large language models](#). In *Proceedings of the 1st Workshop on Personalization of*

- Generative AI Systems (PERSONALIZE 2024)*, pages 1–8. Association for Computational Linguistics.
- Yu-Min Tseng, Yu-Chao Huang, Teng-Yun Hsiao, Wei-Lin Chen, Chao-Wei Huang, Yu Meng, and Yun-Nung Chen. 2024. [Two tales of persona in llms: A survey of role-playing and personalization](#). *Preprint*, arXiv:2406.01171.
- Quan Tu, Shilong Fan, Zihang Tian, Tianhao Shen, Shuo Shang, Xin Gao, and Rui Yan. 2024. [Charactereval: A chinese benchmark for role-playing conversational agent evaluation](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11836–11850, Bangkok, Thailand. Association for Computational Linguistics.
- Jack Urbanek, Angela Fan, Siddharth Karamcheti, Saachi Jain, Samuel Humeau, Emily Dinan, Tim Rocktäschel, Douwe Kiela, Arthur Szlam, and Jason Weston. 2019. [Learning to speak and act in a fantasy text adventure game](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*.
- Junqiao Wang, Kunyu Wu, and Yuqi Ouyang. 2025a. [Enhancing character-coherent role-playing dialogue with a verifiable emotion reward](#). *Information*, 16(9):738.
- Kai Wang, Haoyang You, Yang Zhang, and Zhongjie Wang. 2026a. [Memory-driven role-playing: Evaluation and enhancement of persona knowledge utilization in llms](#). *Preprint*, arXiv:2603.19313.
- Lei Wang, Jianxun Lian, Yi Huang, Yanqi Dai, Haoxuan Li, Xu Chen, Xing Xie, and Ji-Rong Wen. 2025b. [CharacterBox: Evaluating the role-playing capabilities of LLMs in text-based virtual worlds](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6372–6391, Albuquerque, New Mexico. Association for Computational Linguistics.
- Noah Wang, Zhongyuan Peng, Haoran Que, Jiaheng Liu, Wangchunshu Zhou, Yuhao Wu, Hongcheng Guo, Ruitong Gan, Zehao Ni, Jian Yang, Man Zhang, Zhaoxiang Zhang, Wanli Ouyang, Ke Xu, Wenhao Huang, Jie Fu, and Junran Peng. 2024a. [Rolellm: Benchmarking, eliciting, and enhancing role-playing abilities of large language models](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 14743–14777. Association for Computational Linguistics.
- Xintao Wang, Yunze Xiao, Jen-tse Huang, Siyu Yuan, Rui Xu, Haoran Guo, Quan Tu, Yaying Fei, Ziang Leng, Wei Wang, Jiangjie Chen, Cheng Li, and Yanghua Xiao. 2024b. [Incharacter: Evaluating personality fidelity in role-playing agents through psychological interviews](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1840–1873. Association for Computational Linguistics.
- Xudong Wang, Chaoning Zhang, Chenghao Li, Shuxu Chen, Qigan Sun, Jiaquan Zhang, Fachrina Dewi Puspitasari, Tae-Ho Kim, Jiwei Wei, Malu Zhang, Guoqing Wang, Yang Yang, and Heng Tao Shen. 2026b. [Agent-GWO: Collaborative agents for dynamic prompt optimization in large language models](#). *Preprint*, arXiv:2604.18612.
- Xudong Wang, Chaoning Zhang, Qigan Sun, Zhenzhen Huang, Chang Lu, Sheng Zheng, Zeyu Ma, Caiyan Qin, Yang Yang, and Hengtao Shen. 2026c. [Transforming external knowledge into triplets for enhanced retrieval in RAG of LLMs](#). *Preprint*, arXiv:2604.12610.
- Ye Wang, Jiaxing Chen, and Hongjiang Xiao. 2026d. [Role-playing agents driven by large language models: Current status, challenges, and future trends](#). *Preprint*, arXiv:2601.10122.
- Yongjie Wang, Jonathan Leung, and Zhiqi Shen. 2025c. [Rolerag: Enhancing llm role-playing via graph guided retrieval](#). *Preprint*, arXiv:2505.18541.
- Zongsheng Wang, Kaili Sun, Bowen Wu, Qun Yu, Ying Li, and Baoxun Wang. 2025d. [Raiden-r1: Improving role-awareness of llms via grpo with verifiable reward](#). *Preprint*, arXiv:2505.10218.
- Chupeng Wei, Zheng Chen, Jie Zhang, and Jia Liu. 2026. [Roleplayllm: Enhancing role-playing abilities in large language models via multidimensional fine-tuning and preference optimization](#). In *Natural Language Processing and Chinese Computing*, volume 16104 of *Lecture Notes in Computer Science*. Springer.
- Yangbo Wei, Zhen Huang, Fangzhou Zhao, Qi Feng, and Wei W. Xing. 2025. [MECoT: Markov emotional chain-of-thought for personality-consistent role-playing](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 8297–8314. Association for Computational Linguistics.
- Weihao Wu, Liang Cao, Xinyu Wu, Zhiwei Lin, Rui Niu, Jingbei Li, and Zhiyong Wu. 2025. [Voxrole: A comprehensive benchmark for evaluating speech-based role-playing agents](#). *Preprint*, arXiv:2509.03940.
- Rui Xu, Mingyu Wang, Xintao Wang, Dakuan Lu, Xiaoyu Tan, Wei Chu, and Yinghui Xu. 2025. [Guess what I am thinking: A benchmark for inner thought reasoning of role-playing language agents](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 15148–15168, Suzhou, China. Association for Computational Linguistics.
- Bohao Yang, Dong Liu, Chen Tang, Chenghao Xiao, Kun Zhao, Chao Li, Lin Yuan, Guang Yang, Lanxiao Huang, and Chenghua Lin. 2024. [Crafting customisable characters with llms: Introducing simschat, a persona-driven role-playing agent framework](#). *Preprint*, arXiv:2406.17962.

- Jisoo Yang, Jongwon Ryu, Minuk Ma, Trung X. Pham, and Junyeong Kim. 2026a. [The pragmatic persona: Discovering llm persona through bridging inference](#). *Preprint*, arXiv:2604.24079.
- Shihao Yang, Zhicong Lu, Yong Yang, Bo Lv, Yang Shen, and Nayu Liu. 2026b. [HyCoRA: Hyper-contrastive role-adaptive learning for role-playing](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 34304–34312.
- Bingsheng Yao, Bo Sun, Yuanzhe Dong, Yuxuan Lu, and Dakuo Wang. 2025. [Dprf: A generalizable dynamic persona refinement framework for optimizing behavior alignment between personalized llm role-playing agents and humans](#). *Preprint*, arXiv:2510.14205.
- Jiaquan Zhang, Chaoning Zhang, Shuxu Chen, Zhenzhen Huang, Pengcheng Zheng, Zhicheng Wang, Ping Guo, Fan Mo, Sung-Ho Bae, Jie Zou, Jiwei Wei, and Yang Yang. 2026. [Lightweight LLM agent memory with small language models](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12914–12929, San Diego, California, United States. Association for Computational Linguistics.
- Saizheng Zhang, Emily Dinan, Jack Urbanek, Arthur Szlam, Douwe Kiela, and Jason Weston. 2018. [Personalizing dialogue agents: I have a dog, do you have pets too?](#) In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*.
- Wenyuan Zhang, Shuaiyi Nie, Jiawei Sheng, Zefeng Zhang, Xinghua Zhang, Yongquan He, and Tingwen Liu. 2025a. [Revealing and mitigating the challenge of detecting character knowledge errors in LLM role-playing](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 33279–33302, Suzhou, China. Association for Computational Linguistics.
- Xiaotian Zhang, Ruizhe Chen, Yang Feng, and Zuozhu Liu. 2025b. [Persona-judge: Personalized alignment of large language models via token-level self-judgment](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 5037–5049. Association for Computational Linguistics.
- Xueqiao Zhang, Chao Zhang, Jingtao Xu, Yifan Zhu, Xin Shi, Yi Yang, and Yawei Luo. 2025c. [Video2roleplay: A multimodal dataset and framework for video-guided role-playing agents](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 23677–23703, Suzhou, China. Association for Computational Linguistics.
- Jinfeng Zhou, Yongkang Huang, Bosi Wen, Guanqun Bi, Yuxuan Chen, Pei Ke, Zhuang Chen, Xiyao Xiao, Libiao Peng, Kuntian Tang, Rongsheng Zhang, Le Zhang, Tangjie Lv, Zhipeng Hu, Hongning Wang, and Minlie Huang. 2025. [Characterbench: Benchmarking character customization of large language models](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 26101–26110.
- Junkai Zhou, Liang Pang, Huawei Shen, and Xueqi Cheng. 2023. [Simoap: Improve coherence and consistency in persona-based dialogue generation via over-sampling and post-evaluation](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9945–9959, Toronto, Canada. Association for Computational Linguistics.
- Yitian Zhou, Chaoning Zhang, Jiaquan Zhang, Zhenzhen Huang, Jinyu Guo, Sung-Ho Bae, Lik-Hang Lee, Caiyan Qin, and Yang Yang. 2026. [From similarity to structure: Training-free LLM context compression with hybrid graph priors](#). *Preprint*, arXiv:2604.23277.

A Comparison with Related Surveys

The following comparison highlights how existing surveys differ in their main scopes, methodologies, and treatments of consistency, and clarifies how our work complements them.

- Broad role-playing with language models, including data, models, agents, and applications** (Chen et al., 2024b). General taxonomy of RPA components and challenges. *Discusses persona and character consistency as part of the broader RPA landscape. We make consistency the central organizing problem and analyze it as role-state maintenance.*
- Persona, role-playing, and personalization across language agents** (Chen et al., 2024a). Connects persona construction, personalized generation, and evaluation. *Treats consistency as one requirement among persona and personalization goals. We separate role consistency from user personalization and emphasize longitudinal role boundaries.*
- Two uses of persona in LLMs: role-playing and personalization** (Tseng et al., 2024). Clarifies conceptual relations between persona-driven interaction and personalized systems. *Covers persona consistency but not as a full systems-level taxonomy. We extend from persona consistency to identity, memory, reasoning, alignment, and safety consistency.*
- Design guidelines for evaluating LLM-based RPAs** (Chen et al., 2025a). Evaluation design, benchmark construction, and assessment dimensions. *Strongest overlap on evaluation of role-playing quality. We treat evaluation as one layer and connect it to memory, reasoning, training, and boundary-aware alignment.*
- Current status, challenges, and trends of LLM-driven RPAs** (Wang et al., 2026d). Broad overview of architectures, capabilities, applications, and future trends. *Identifies consistency as an important challenge. We provide a narrower taxonomy centered on maintaining coherent role state under long, changing, and adversarial interaction.*

B Word Cloud Keyword List

Figure 2 is based on titles and abstracts from 100 arXiv papers retrieved with role-playing-agent-related queries. The sample was constructed from arXiv search results using queries such as *role-playing agent*, *roleplay agent*, *persona agent large language model*, *character agent large language model*, *persona consistency large language model*, *character consistency large language model*, and *role consistency agent*. We deduplicated papers by arXiv identifier, scored candidates with positive RPA indicators (e.g., role-playing, persona, character, in-character, dialogue agent, virtual character, NPC, and LLM) and negative filters for unrelated uses of RPA (e.g., robotic process automation and remote piloted aircraft), then retained the top 100 papers whose relevance score exceeded the filtering threshold.

Keyword hotness was computed from each paper’s title and abstract. We first normalized variants into canonical phrases, including role-playing, persona, character, memory, retrieval, alignment, safety, reasoning, multimodal, personality, style, identity, knowledge, dialogue, multi-turn, simulation, game, training, preference optimization, and benchmark. Remaining unigram candidates were lowercased, stripped of punctuation, filtered with stopwords, and kept only if they appeared at least three times. For each term k , we computed term frequency tf_k , document frequency df_k , inverse document frequency

$$idf_k = \log \frac{1 + N}{1 + df_k} + 1, \quad (1)$$

where $N = 100$. To emphasize recent activity, we also computed

$$growth_k = \log(1 + tf_k^{2025+}) - \log(1 + tf_k^{<2025}), \quad (2)$$

and ranked terms by

$$hotness_k = tf_k \cdot idf_k \cdot r_k \cdot c_k, \quad (3)$$

where $r_k = 1 + 0.25 \max(growth_k, 0)$ is the recency factor and $c_k = 1 + 0.2 \min(df_k/N, 1)$ is the document-coverage factor. For the rendered word cloud, generic search-anchor terms such as *role-playing* and *large language model* were removed, and consistency-family terms, including *consistency*, *consistent*, *persona consistency*, *character consistency*, *role consistency*, *fidelity*, and *coherence*, were merged into *consistency*. Table 3

lists the complete keyword set used in the rendered word cloud. Table 2 below reports selected unmerged diagnostic signals for interpretability, so its values should not be directly compared with merged word-cloud entries such as *consistency*.

C Additional Sample Diagnostics

To make the descriptive analysis behind Figure 2 more explicit, we add two lightweight diagnostics over the same 100-paper arXiv sample. These diagnostics are not intended as exhaustive bibliometric evidence; rather, they provide a transparent check on the topical shape of the sample used to motivate this survey.

Figure 7 shows that the sampled RPA consistency literature is strongly recent: 74 of the 100 papers were submitted in 2025 or 2026. This concentration supports the motivation for a focused survey, but it also explains why the literature remains methodologically unsettled: many papers are preprints, benchmark definitions are still changing, and several evaluation dimensions have not yet stabilized into common protocols.

The right panel of Figure 7 maps titles and abstracts to the six consistency dimensions used in our benchmark map: knowledge/factual consistency (K), expression/style consistency (E), psychological/personality fidelity (P), memory and temporal continuity (M), role-aware reasoning and decision consistency (R), and robustness to role drift or boundary pressure (B). Personality, expression, and memory signals appear in roughly half of the sample, while reasoning and boundary robustness appear less often. This pattern is consistent with the main-text argument: current work increasingly recognizes that RPAs need more than factual persona matching, but evaluation and methods for decision-level consistency and adversarial boundary control are still comparatively sparse.

Table 2 further clarifies which mechanisms are prominent in the sample. Personality, reasoning, training, alignment, and benchmarks are among the strongest signals, suggesting that the field is moving from static character description toward controllable behavior and evaluation. At the same time, safety, multimodality, multi-turn continuity, and preference optimization have high recent concentration but smaller paper coverage. We therefore treat these areas as future-facing rather than settled: they motivate the open challenges in Section 6, but they should not yet be interpreted as

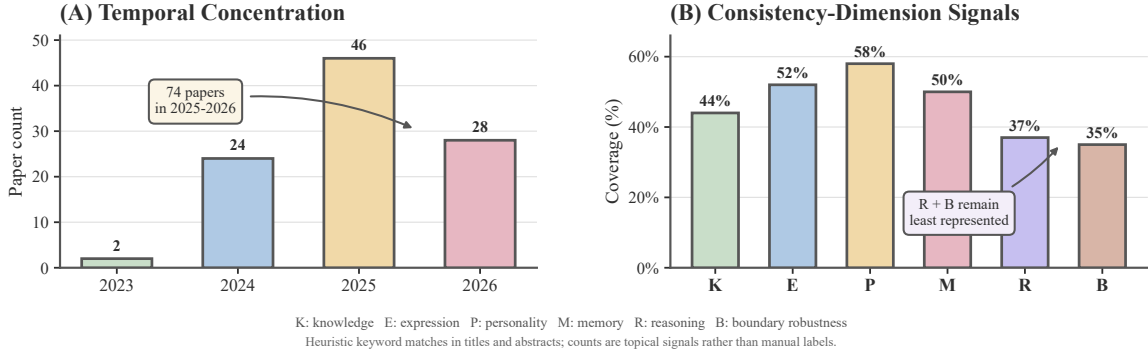


Figure 7: **Additional diagnostics over the 100-paper arXiv sample.** Left: submitted-year distribution. Right: heuristic title/abstract coverage of the six role-consistency dimensions used in Table 1. Counts are based on keyword matches and should be read as topical salience indicators, not as manual labels.

Keyword signal	Hotness	TF	Papers	Recent TF	Earlier TF	Interpretation
Recent TF = 2025–2026 term frequency; Earlier TF = pre-2025 term frequency.						
Role-state content and behavior						
personality	341.163	104	25	124	33	trait and psychological fidelity are highly visible
reasoning	242.400	65	28	107	10	role-aware deliberation is a fast-growing topic
consistency	168.331	40	27	39	1	explicit consistency framing is mostly recent
memory	108.231	29	11	36	18	memory is visible but fragmented across systems
style	107.419	22	9	35	5	expressive consistency is often implicit
knowledge	89.157	26	16	36	16	factual role knowledge remains a baseline concern
multi-turn	65.097	18	13	22	10	long-horizon interaction is not yet universal
Methods, resources, and emerging constraints						
training	190.348	61	34	79	14	supervised and alignment data remain central
alignment	165.991	44	29	68	5	role fidelity is increasingly tied to optimization
benchmark	163.108	58	31	42	16	evaluation resources are a major research driver
multimodal	151.114	22	7	38	0	multimodal RPAs are an emerging frontier
safety	144.922	21	6	30	0	safety and boundary control remain underrepresented
retrieval	110.363	32	9	22	20	retrieval is present but less dominant than training
preference optimization	18.318	3	2	3	0	preference methods are still early in RPA-specific work

Table 2: **Selected keyword diagnostics from the 100-paper arXiv sample.** Hotness follows the scoring formula in Appendix B; TF denotes term frequency in titles and abstracts. These diagnostic rows use unmerged keyword signals, while Table 3 reports the merged keyword list used for the rendered word cloud.

mature subfields with standardized tasks and metrics.

How to read the hotness scores. The ranking is designed to organize the sampled literature, not to estimate the absolute importance of a topic. A high score can arise from repeated mentions within papers, broad document coverage, recent growth, or a combination of these factors. Consequently, the score should be interpreted comparatively within this sample rather than as a calibrated measure across corpora. The merged *consistency* entry is a particularly important example: it pools several lexical variants so that the central construct is not artificially split across near-synonyms. Its magnitude therefore indicates both topical centrality and normalization choices, whereas lower-ranked entries may represent narrower concepts expressed with less standardized terminology.

What the leading terms suggest. The head of the ranking combines role descriptors, such as *character*, *persona*, and *personality*, with process-oriented terms, such as *reasoning*, *interaction*, *training*, and *alignment*. This mixture captures a transition in the literature. Earlier work often treated the role primarily as a profile to be imitated; more recent work increasingly studies how that profile is maintained through decisions, dialogue history, optimization, and environmental feedback. The prominence of *benchmark* further shows that evaluation resources have become part of the research problem itself. These signals support the layered view developed in the main paper: role consistency depends not only on what an agent says, but also on how it updates state, selects actions, handles pressure, and is measured over time.

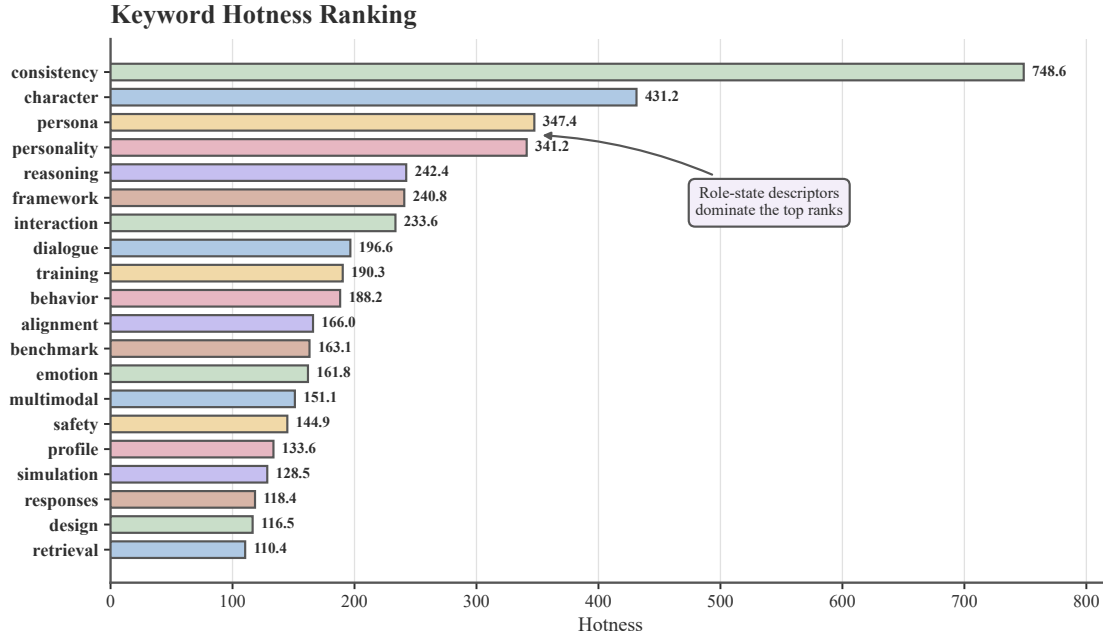


Figure 8: **Keyword hotness ranking for the 100-paper arXiv sample.** The bar chart provides a quantitative companion to the word cloud in Figure 2.

Reading the head and tail. Figure 8 makes the steepness of the ranking visible. Broad role descriptors form the head, process and capability terms occupy the middle, and more specific mechanisms extend through the tail. Lower rank does not imply low importance: terms such as *memory*, *multi-turn*, *identity*, and *preference optimization* apply to narrower settings and may also be split across related expressions such as state tracking, long-horizon dialogue, or policy optimization. The ranking is therefore most informative when read with the taxonomy and benchmark analysis rather than as a standalone research agenda.

Reproducibility and limits. Table 3 gives the exact ordering behind the word cloud and bar chart; its four column pairs are a compact layout, not semantic groups. Together, the cloud supports rapid

recognition, the chart exposes relative magnitude, and the table preserves rounded scores for reconstruction. These values measure lexical salience within the retrieved sample, not method quality or empirical effectiveness. Titles and abstracts may omit technical distinctions, and the queries intentionally favor role-playing and consistency research. We use the diagnostics only for triangulation and reproducibility; claims about methods and benchmark coverage continue to rely on the paper-level review in the main text.

#	Keyword / Hotness	#	Keyword / Hotness	#	Keyword / Hotness	#	Keyword / Hotness
1	consistency / 748.649	2	character / 431.192	3	persona / 347.439	4	personality / 341.163
5	reasoning / 242.400	6	framework / 240.805	7	interaction / 233.567	8	dialogue / 196.596
9	training / 190.348	10	behavior / 188.224	11	alignment / 165.991	12	benchmark / 163.108
13	emotion / 161.845	14	multimodal / 151.114	15	safety / 144.922	16	profile / 133.550
17	simulation / 128.481	18	responses / 118.432	19	design / 116.473	20	retrieval / 110.363
21	social / 108.971	22	memory / 108.231	23	dynamic / 108.080	24	style / 107.419
25	systems / 106.353	26	multi-agent / 102.540	27	generation / 100.813	28	traits / 99.041
29	scenario / 93.150	30	knowledge / 89.157	31	cognition / 88.194	32	multi-turn / 65.097
33	game / 53.404	34	world / 44.071	35	identity / 40.846	36	role-aware / 37.943
37	virtual / 37.647	38	charactereval / 27.207	39	characterbox / 24.659	40	preference opt. / 18.318

Table 3: **Complete keyword list used to render the word cloud in Figure 2.** Hotness scores are rounded to three decimals.